



**Verified Carbon
Standard**

INSTALLATION OF HIGH EFFICIENCY WOOD BURNING COOKSTOVES IN PHILIPPINES

Document Prepared by

C-Quest Capital Stoves Asia Limited

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Prepared By	C-Quest Capital Stoves Asia Limited
Contact	Address: C-Quest Capital Stoves Asia Limited, Brumby Centre, Lot 42, Jalan Muhibbah, 87000 Labuan F. T., Malaysia, Telephone: +6 087 593828 Email: cgc-operations@cquestcapital.com Website: www.cquestcapital.com

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1 PROJECT DETAILS

1.1 Summary Description of the Project

A summary description of the technologies/ measures to be implemented by the project.

The project involves distribution of fuel-efficient improved cookstoves (ICS) in Philippines. The ICS disseminated through this project will replace the baseline cookstoves.

Through this project, the distribution and installation of approximately 1,000,000 ICS will be undertaken for households in Philippines. It is intended that under this project two types of single pot portable stoves will be distributed, which are SSM S32-X stove and Baldwin stove, or any other similar ICS model may be added based on the requirement of end users as suitable. Each household will receive up to 2 ICS, it could comprise of different types of ICS¹. The ICS will burn wood more efficiently thereby improving thermal transfer to pots, hence saving fuel. This not only will halt the rapid deforestation in Philippines but will also reduce health hazards from indoor air pollution and women and children will have to spend less time collecting firewood.

The end user will be informed in advance that the use of ICS generates carbon credits which in turn will be used for subsidising the price of ICS and for recovering project implementation costs.

The location of the project.

The project will take place in Philippines. The details of the project location are provided in Section 1.12.

An explanation of how the project is expected to generate GHG emission reductions or removals.

The ICS will substitute the current cooking practices of using the three stone fired stoves or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. The ICS burns wood more efficiently thereby improving thermal transfer to pots, hence saving fuel and lowering greenhouse gas emissions.

¹ CQC has implemented ICS projects with single ICS for HHs in its previous projects (PoA 9007, PoA 9558, PoA 8060), where it was observed that a single stove is insufficient for the household requirements and the end users continued to use the traditional stoves along with the project ICS. This led to a more practical and sustainable two-stove distribution initiative per household by CQC. This approach will ensure that the end users have complete access to clean cooking solutions and that the devices are sufficient for their cooking requirements, ensuring that the household doesn't go back to the baseline practice of three-stone fired/tripod stoves.

Even recent projects implemented by CQC (VCS 2521, 2522, 2523) have adopted the same approach followed by the current project activity too.

A brief description of the scenario existing prior to the implementation of the project.

The baseline scenario is the continued use of three stone fired or a conventional cook stove without a grate or a chimney by the target population burning non-renewable wood fuel to meet similar thermal energy needs in the absence of project activity.

An estimate of annual average and total GHG emission reductions and removals.

Total GHG emission reduction from the project is expected to be 25,903,615 tCO₂e over the entire crediting period of 10 years. The average annual GHG emission reduction from the project is expected to be 2,590,361 t CO₂e.

Audit Type	Period	Program	VVB Name	Number of years
Validation	20-June-2023 ²	VCS	Earthood Services Private Limited	0.71 Years (257 Days) ³

1.2 Sectoral Scope and Project Type

The project is categorised under type/category as below:

- a) **Sectoral scope**: 03 - Energy demand
- b) **Type**: II – Energy efficiency improvement projects

The project is a grouped project.

1.3 Project Eligibility

The project involves energy efficient cookstove distribution which falls under the category of efficiency improvements in thermal applications, therefore it is eligible under the scope of VCS Program.

Applicability of VCS Programme scope as per VCS standard Ver 4.4 para 2.1 as follows:

Scope as per para 2.1.1 of VCS standard V4.4	Applicability
1 The six Kyoto Protocol greenhouse gases	The project activity reduces GHG emissions associated with the combustion of cooking fuels. Gases included are CO ₂ , CH ₄ , N ₂ O.

² Date of validation report by VVB .

³ Number of years is duration between appointment of VVB (07-October-2022) and Validation report (20-June-2023) which is less than a year

2 Ozone-depleting substances.	The project does not involve any ozone depleting substances as project devices use fuelwood in the ICS.
3 Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	The ICS distributed (& to be distributed) under the project activity uses the methodology “VMR0006 “Methodology for Installation of High Efficiency Firewood Cookstoves”, version 1.1.” which is a VCS approved methodology
4 Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).	Not applicable
5 Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements	Not applicable
Scope as per para 2.1.2 of VCS standard V4.4	Applicability
The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.	Not applicable as the project is voluntary and GHG emissions are not for the purpose of their subsequent reduction, removal or destruction
Scope as per para 2.1.2 of VCS standard V4.4	Applicability
The VCS Program also excludes the project activities under the circumstances indicated in Table 1, of VCS standard 4.4	Also, the project activity is <i>Installation of High Efficiency Firewood Cookstoves</i> is not listed in the category of excluded projects in Table 1 of item 2.1.3 of VCS Standard ver. 4.4, and hence, project activity is eligible under the scope of the VCS Program.

The Project Activity falls within the scope of the VCS Program as per VCS standard V 4.4 para 2.1.

Compliance with other general requirements are as described below:

Requirements as per para 3.1 of VCS Standard v4.4	Applicability
3.1.1 Projects shall meet all applicable rules and requirements set out under the VCS Program, including this document. Projects shall be guided by the principles set out in Section 2.2.1.	The project meets all applicable rules and requirements set out under the VCS program as outlined in this section above table. The project applies verra approved methodology (refer section 3.1 & 3.2 of this PD)all major emissions have been clearly identified (refer section 3.3), PP has used conservative assumptions wherever necessary, maintained transparency and accuracy in disclosing and consistently reported wherever necessary.
3.1.2 Projects shall apply methodologies eligible under the VCS Program. Methodologies shall be applied in full, including the full application of any tools or modules referred to by a methodology, noting the exception set out in Section 3.14.1. The list of methodologies and their validity periods is available on the Verra website.	Project has applied VCS approved methodology VMR0006 ver1.1, along with the tools or modules as applicable (details provided in section 3.1 of this PD) .
3.1.3 Projects shall apply the latest version of the applicable methodology in all cases unless a grace period applies to the project as set out in 3.21 below. Projects must update to the latest version of the methodology when reassessing the baseline and renewing a crediting period.	Project has applied VCS approved methodology VMR0006 V1.1 ⁴ . Refer to section 3 of the PD for more details.
3.1.4 Projects and the implementation of project activities shall not lead to the violation of any applicable law, regardless of whether or not the law is enforced.	The project and its instances shall remain in compliance with the applicable local laws, The project implementation does not lead to violation of any applicable law(s) (section 1.14 of this document be referred for details).
3.1.5 Where projects apply methodologies that permit the project proponent its own	The project applies the chosen formulae as outlined in the applied methodology VMR00006

⁴ The grace period for listed projects to complete validation under VMR0006, v1.1 ended on 28-June 2023. The project has been submitted to VERRA for request for registration on 26 -June-2023.

choice of model (see the VCS Program document VCS Program Definitions for the definition of model), the model shall meet the requirements set out in the VCS Program document VCS Methodology Requirements, and it shall be demonstrated at validation that the model is appropriate to the project circumstances (i.e., use of the model will lead to an appropriate quantification of GHG emission reductions or removals).	V1.1 and details of formulas used are provided in Sec4.4 of this PD. As the model is not at PP choice of discretion, the same not applicable.
3.1.6 Where projects apply methodologies that permit the project proponent to choose a third-party default factor or standard to ascertain GHG emission data and any supporting data for establishing baseline scenarios and demonstrating additionality, such default factor or standard shall meet the requirements set out in the VCS Program document VCS Methodology Requirements.	Not applicable, the project does not apply any third-party default factor.
3.1.7 Where the rules and requirements under an approved GHG program conflict with the rules and requirements of the VCS Program, the rules and requirements of the VCS Program shall take precedence.	The project will follow VCS rules and requirements.
3.1.8 Where projects apply methodologies from approved GHG programs, they shall conform with any specified capacity limits (see the VCS Program document VCS Program Definitions for the definition of capacity limit) and any other relevant requirements set out with respect to the application of the methodology and/or tools referenced by the methodology under those programs.	Not applicable as project has applied only VCS approved methodology (VMR0006 ver1.1)
3.1.9 Where Verra issues new VCS Program rules, the effective dates of these requirements are set out in Appendix 3	The project will follow new VCS program rules with effective dates as and when applicable as per Appendix 3 of VCS standard Version 4.4.

Document History and Effective Dates or equivalent for other program documents and are listed in a companion Summary of Effective Dates document which corresponds with each update.	
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1.4 Project Design

The project is a grouped project.

Eligibility Criteria

For the inclusion of new project activity instances (each SSM-S32-X Stove or Baldwin Stove), the project proponent shall ensure that it meets the eligibility criteria below.

No.	Criterion	Compliance
1	Meet the applicability conditions set out in the methodology applied to the project	New project activity instances (each SSM-S32-X Stove or Baldwin Stove) will meet the applicability conditions set out in Section 3.2 where the target of the end-user is household and the ICS deployed is at least 25% of thermal efficiency.
2	Use the technologies or measures specified in the project description.	SSM-S32-X Stoves and Baldwin Stoves to be adopted in the project,
3	Apply the technologies or measures in the same manner as specified in the project description.	SSM-S32-X Stoves and Baldwin Stoves to be adopted in the project and it will replace traditional cookstoves in household
4	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	The new project activity instances will be distributed within Philippines only and subject to the same baseline scenario determined in Section 3.4.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	All new project activity instances will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no mandated government programme or policy in host country of this project ensuring the distribution of new project activity instances. Step 2: Positive List The inclusion of new project activity instances will comply with positive list as it satisfies criterion (1) of the positive list of applied methodology where it meets all the applicability conditions of the methodology.

6	Where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit. Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows: 1) Each project activity instance that exceeds one percent of the capacity limit shall be identified. 2) Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometer of at least one other instance in the cluster. Instances that are not within one kilometer of any other instance shall not be assigned to clusters. 3) None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit.	No project activity instance will exceed the applicable limit, which is 180 GWh_{th}/y. The expected annual energy saving for each project activity instance is approximately 0.02 GWh_{th}/y or 0.011% of the limit. As the annual energy saving is below 1% of the limit, no project activity instance is identified and divided into clusters.
7	Double counting: The Improved Cookstove distributed under this project shall be uniquely identifiable based on the distribution records.	Project Activity Instance Database containing unique serial number of each ICS. Each ICS distributed will have corresponding end user (Stove owner) details (i.e., name, geographical coordinates, address, unique serial number). The combination of this unique serial number of each ICS along with geographical address of the stove owner confirms the unique identity of each project device
8	Target Population: End users for each project activity instance shall be households, with non-renewable biomass on inefficient firewood stoves.	Each ICS will be assigned a unique identification (serial number/alpha code) with name of ICS user, address, Geographical area of distribution, stove model, distribution date, etc.,
9	item 3.6.8 of VCS standard 4.4 The project proponent shall include in a singular project all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent (i.e., instances of the same project activity may not be spread across more than one project if they are within ten kilometers of each other).	As per VCS standard Ver 4.4 Appendix 3 " Updated language in Sections 3.6.4 – 3.6.17 to provide added clarity around requirements and eligibility criteria for grouped projects, projects with multiple activity instances, and capacity limits. For projects that request listing after 21 December 2022, this update is effective immediately.

		For projects that requested listing on or before 21 December 2022: (i). These requirements do not apply if validation is completed by 30 June 2023 As of the grouped project activity under VCS listed on 19-August-2022 and completed validation on 20-June-2023 and submitted for registration on 26-June-2023. So these additional requirements not applicable for the grouped project activity.
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Along with the above points, the Inclusion of New Project Activity Instances⁵ follows the below criteria (as per Sec 3.6.17 of VCS standard V4.4)

No.	Requirements as per para 3. 6.17 of VCS Standard v4.4	Compliance
1	Occur within one of the designated geographic areas specified in the project description	New project activity instances (Energy Efficient Cook Stoves) will be distributed in the designated geographic areas specified in the project description.
2	Conform with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial conformance with multiple sets of eligibility criteria is insufficient.	New project activity instances will comply with a complete set of eligibility criteria as mentioned above for the inclusion of new project activity instances
3	Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	New project activity instances information to be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the VVB.
4	Be included in an updated project description, with updated project location information (as set out in Section 3.11), which shall be validated at the time of verification against the applicable set of eligibility criteria	New project activity instances should be included by validating eligibility criteria as mentioned in this PD .
5	Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance	A sample beneficiary agreement with end users including the provision that emission reductions generated by the project activity are owned by the Project Proponents will be provided for project activity instance.

⁵ Grouped projects provide for the inclusion of new project activity instances subsequent to the initial validation of the project

	began reducing or removing GHG emissions).	
6	Have a start date that is the same as or later than the grouped project start date.	New project activity instances' start date should be after the start date of grouped project activity
7	Be eligible for crediting from the start date of the project activity instance through to the end of the project crediting period (only).	New project activity instances are eligible to claim credits from crediting period start date to the end of the project crediting period.
8-	Only eligible for crediting from the start of the verification period in which they were added to the grouped project.	As the updated language in VCS Standard v4.4, Sections 3.6.4 – 3.6.17 are not applicable for the project those have been listed on or before 21 December 2022 and validation is completed by 30 June 2023. The current project was listed on 19-August 2023 and validation was completed on 20-June-2023. Therefore, the revised condition mentioned in section 3.6.8 – 3.6.17 of VCS standard v4.4 are not applicable to the current project. Applicability for Section 3.6.9 for the grouped project activity , already provided in the above table in this section (vide sr.no 6.)
9	Not be or have been enrolled in another VCS project.	
10	Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 - 3.6.9.	

S No.	Requirements as per para 3.6.10 to 3.6.15 as per VCS standard V 4.4	Applicability
1	3.6.10 Grouped projects shall specify one or more clearly defined geographic areas within which project activity instances may be developed.	All project activity instances under grouped project activity located within the geographical area within the host country of Philippines (Details provided in Sec 1.12 of this PD)
2	3.6.11 Determination of baseline scenario and demonstration of additionality are based upon the initial project activity instances.	The baseline scenario for each project technology and geographic area, as defined in section 3.4 of the VCS PD, is applicable to the corresponding new project activity instances under the specific technology, PP will replace the traditional three stone fire cookstove with the improved efficient cook stove, where the usage of the firewood would be reduced by the ICS. In addition, that each new project activity instance included within the grouped project follows the additionality as per sec 3.5 of this VCS PD.
3	3.6.12 As with non-grouped projects, grouped projects may incorporate multiple project activities (see Section	Not applicable as the grouped project activity does not includes more than one

	3.6.1 – 3.6.3 for more information on multiple project activities). Where a grouped project includes multiple project activities, the project description shall designate which project activities may occur in each geographic area.	project activity and/or where projects apply more than one methodology. Grouped project activity applies only VCS approved methodology VMR0006 V1.1 as outlined in Sec 3.1 of the PD.
4	3.6.13 The baseline scenario for a project activity shall be determined for each designated geographic area, in accordance with the methodology applied to the project	The baseline scenario for a project activity be determined for each designated geographic area as can be known from each end user registrations which has the details of the Unique serial number/alpha code, about the baseline stove too.
5	3.6.14 The additionality of the initial project activity instances shall be demonstrated for each designated geographic area, in accordance with the methodology applied to the project.	Additionality of the of the grouped project activity has been described in sec 3.5 of this PD and where in the geographical area is the host country Philippines.
6	3.6.15 Where factors relevant to the determination of the baseline scenario or demonstration of additionality require assessment across a given area, the area shall be, at a minimum, the grouped project geographic area. Examples of such factors include, inter alia, common practice; laws, statutes, regulatory frameworks, or policies relevant to demonstration of regulatory surplus; determination of regional grid emission factors; and historical deforestation and degradation rate	The project boundary is the Philippines as detailed in sec 1.12 and the additionality assessment has considered the host country Philippines as geographic area for baseline or demonstration of additionality .

1.5 Project Proponent

Organization name	C-Quest Capital Stoves Asia Limited
Contact person	Ken Newcombe
Title	Director
Address	Brumby Centre, Lot 42, Jalan Muhibbah, 87000 Labuan, Malaysia.
Telephone	+6 087 423828
Email	cqc-operations@cquestcapital.com

1.6 Other Entities Involved in the Project

Organization name	N/A
Contact person	N/A
Title	N/A
Address	N/A
Telephone	N/A
Email	N/A

1.7 Ownership

The project ownership is with C-Quest Capital Stoves Asia Limited.

Further, Registration certificate cum consent deed signed by end user (stove owner) to PP during the registration process includes the statutory clause *“Acknowledgement and Carbon Rights Waiver – My household hereby transfers and waives all and any rights in connection with the improved cookstove or project and greenhouse gas emission reductions or other environmental or social attributes generated by the installation and use of the improved cookstove to C-Quest Capital Stoves Asia Ltd and we understand that C-Quest Capital Stoves Asia Ltd may, at any time, transfer those rights to an affiliate without further notice or consent”*.

This is in line with clause 5 of section 3.7.1. of VCS project standard version 04.4.

1.8 Project Start Date

15-February-2023 (The date of distribution of first ICS under this grouped project)

1.9 Project Crediting Period

15-February-2023 to 14-February-2033, 10 years fixed crediting period

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	
Large project	√

Year	Estimated GHG emission reductions or removals (tCO _{2e})
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15-February 2023 to 14-February- 2024	4,894,396
15-February-2024 to 14-Febrary-2025	4,348,448
15-Febrary-2025 to 14-Febrary-2026	3,815,559
15-February-2026 to 14-Febrary-2027	3,295,538
15-Febrary-2027 to 14-Febrary-2028	2,788,193
15-February-2028 to 14-February-2029	2,293,337
15-February-2029 to 14-Febrary-2030	1,810,785
15-Febrary-2030 to 14-Febrary-2031	1,340,355
15-Febrary-2031 to 14-Febrary-2032	881,865
15-February-2032 to 14-Febrary-2033	435,139
Total estimated ERs	25,903,615
Total number of crediting years	10
Average annual ERs	2,590,361

1.11 Description of the Project Activity

The project involves distribution of fuel-efficient improved cookstoves (ICS) to replace the baseline cookstoves in households. The ICS to be deployed under this project are SSM S32-X stove and Baldwin stove or any other suitable models meeting the applicability criteria which substantially reduce fuel consumption and emissions for conducting cooking activities in households.



Figure1; SSM S32-X stove with pot skirt

Table 1: Technical description of SSM S32-X stove with pot skirt

No	Parameter	Value
1	Thermal Efficiency	51.5% (with pot skirt)
2	Dimensions	25.5 cm height, 32 cm diameter
3	Material	- Stove Body: Stainless Steel - Pot Rest: Cast Iron - Pot Skirt: Stainless Steel - Burning Chamber: Stainless Steel and Ceramic - Fuel Feeding Door: Stainless Steel - Inner Fuel Shelf: Stainless Steel - Outer Fuel Shelf: Powder-coated steel
4	Life Span	10 years

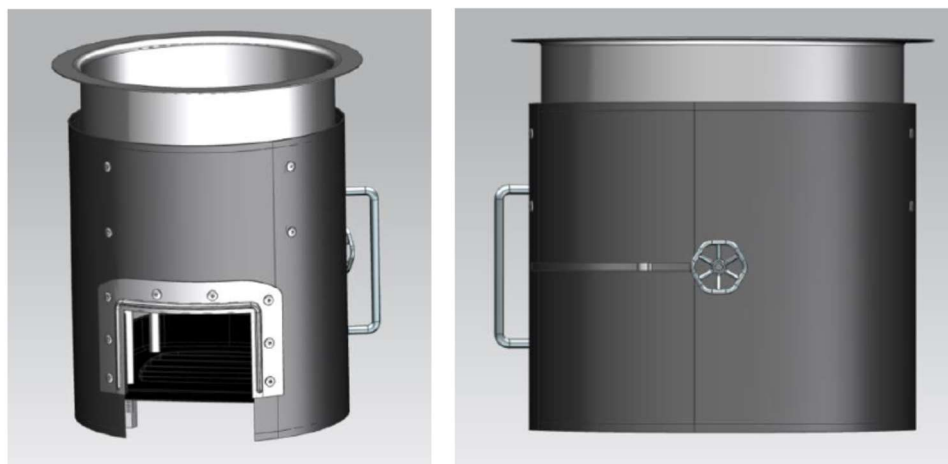


Figure 2: Baldwin stove

Table 2: Technical description of Baldwin stove

No	Parameter	Value
1	Thermal Efficiency	45.0%
2	Dimensions	27.0 cm height; 23.0 cm to 30.0 cm adjustable diameter
3	Material	- Stove Body: Stainless Steel - Pot Rest: Cast Iron - Door frame: Stainless Steel - Pot Rest: Steel - Legs: Steel - Inner Fuel Shelf: Cast Iron
4	Life Span ⁶	10 years

Data collection of ICS end-user

⁶ If the life span of project devices is less than the crediting period, it shall be demonstrated that the devices shall be taken for maintenance/repaired/replaced after the life span has ended. In such cases, if it cannot be demonstrated that the project devices will be replaced with new devices, no emission reductions can be claimed beyond the life span of the project devices.

Project proponents must gather the necessary information to identify households using its ICS during the course of the project. To facilitate this process, each ICS will be assigned a unique serial number. This number will be recorded during the registration process together with the following information (as appropriate and as available):

- Name of ICS user or head of the household
- Geographic area of distribution or GPS of ICS household
- Address ,Phone number of ICS user or household, where available.
- Stove model
- Date of distribution/installation
- ICS serial number
- Retailer/distributor information

The information collected will be stored in the electronic database which will serve as project database for project monitoring and sampling purposes.

1.12 Project Location

The project location will be the geographical boundary of Republic of Philippines with coordinates 12°52'47" N latitude and 121°46.441' E longitude.⁷

⁷ <https://www.geodatos.net/en/coordinates/philippines>



Figure 3: Map of the project area (Republic of Philippines map⁸)

⁸ <https://en.populationdata.net/maps/philippines-regions-and-provinces/>



Figure 4 : Map of project area

1.13 Conditions Prior to Project Initiation

The pre-project scenario is the continued use of three stone fired cookstove or a conventional device with no improved combustion air supply or flue gas ventilation (that is without a grate or a chimney) by the target population by burning non-renewable woody biomass to meet the thermal energy needs. Also, more information is provided in section 3.4 below.

It is a common and traditional practice in the project area to use three stone fired stoves or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney. Hence, the applicable efficiency fraction for three stone fired or conventional stoves which stands at 0.1 (default) as per the applied methodology⁹ is considered in the emission reduction estimations.

The baseline scenario can also be verified from the registration cum consent deed signed by the registered stove user at the time of distribution of the project device (ICS), which confirms that the registered user voluntarily adopted the project stove and was using a three stone fire or conventional device (without a grate or a chimney) before switching to the project stove.

⁹ Pg.19, section 9.2, VMRO006, Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

There are no laws and regulations governing the use of improved cookstoves in Philippines households. The project is a voluntary effort by the project proponent.

A review is made on Philippines environmental laws and regulations¹⁰ as below:

- a) Republic Act No. 9729 s.2009 – Climate Change Act of 2009
- b) Republic Act No. 10587 Environmental Planning Act of 2013
- c) Presidential Decree No. 1586 – Philippine Environmental Impact Statement Law
- d) Presidential Decree No. 1151 – Philippine Environmental Policy
- e) Republic Act no.8749-Philippines Air Control Pollution control policy

There is no negative concern made on improved cookstoves project from the above laws and regulations.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project has not been registered, nor is it seeking registration under any other GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project is not included in an emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit.

Supply Chain (Scope 3) Emissions

As per Section 3.23.9 of the VCS Standard, v4.4, the “producer(s) or retailer(s) of the impacted good or service are known but not involved in the project or do not have a website”. The Project Proponent informed the manufacturers of ICS as well as the implementing partner in Philippines

¹⁰<https://greendevsolutions.com/environmental-laws-and-policies-in-the-philippines/>

at the time of signing agreement for the goods and services provided by them, that ICS will be distributed in Republic of Philippines under this project activity. Also, intimated them about the project registration under VERRA for the generation of VCUs.

A public statement(s) by the Project Proponent has been made throughout the project crediting period on its website for reference. The impacted good or service's producer(s) have been notified of the project and the potential risk of Scope 3 emissions double claiming via email. Evidence of the public statement(s) and email(s) will be provided in this report during validation process.

1.17 Sustainable Development Contributions

The project contributes to sustainable development in a number of ways:

a) Environmental Sustainability

- The project will help significantly reduce greenhouse gas emissions over its lifetime.
- The project will help reduce the use of non-renewable woody biomass from forests, thus will assist in conserving existing forest stock and the protection of natural forest eco-systems and wildlife habitats.

b) Social Sustainability

- The project stoves will considerably reduce the time that need to be spent in collecting wood fuel and cooking activities of the household thereby reducing the work burden on rural families and presenting alternative opportunities for economic development.
- The amount of indoor pollutants from the burning of woody biomass in the household will be reduced. Less carbon dioxide, carbon monoxide and particulates will be emitted due to the decrease in total woody biomass burned.
- The project stoves provide a safer method of burning woody biomass for cooking, thereby helping reducing burn injuries, especially for children in the household.

c) Economic Sustainability

- The project will help develop a section of the local economy, in the distribution, local assembly, maintenance and monitoring activities.
- Household expenditure on cooking fuel will be reduced by the use of the ICS.
- Saved household labor can be diverted to more productive economic activities.
- The employment opportunity will also equip them with skills which are expected to improve their future employment prospects.

The project listed under SD VISTA¹¹ labeling, so details of SDGs achieved by the project activity are provided in the SD VISTA PD.

¹¹ <https://registry.terra.org/app/projectDetail/SDVISTA/3330>

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage as per applied methodology VMR0006 V1.1.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

Not applicable.

2 SAFEGUARDS

2.1 No Net Harm

The project will only bring positive impacts on environmental and socio-economic aspects as elaborated in Section 1.17. No potential negative environmental or socio-economic impacts have been identified for the project.

2.2 Local Stakeholder Consultation

The overall objective of the consultation process was to disseminate information about the project to stakeholders and obtain their views regarding the project.

The Local Stakeholder meeting was conducted physically on the 14-February-2023 and the invitations were sent out on email and the newspaper advertisements. The meeting notice was published in the local newspaper 'The Philippine Star' on 14-January-2023. An advertisement was also posted on the website related to the meeting. This ensured that the project adhered to the criteria of giving reasonable notice to stakeholders. All the stakeholders were invited to provide comments and suggestions about the project.

A total of 105 participants attended the meeting. CQC representatives briefed about the project and the purpose of the stakeholder consultation and then continued with an informative presentation over the project and followed by a detailed description of the VERRA programme, VCU, SDVista and carbon benefits. All the detailed aspects of the project activity were explained including its objectives, technical details of the project stoves, associated social, economic and environmental benefits. Also, the PP explained about the significance of the carbon credits in the project activity and how the carbon finance helps this project feasible.

Feedback was received from several stakeholders. In summary, there was positive feedback regarding the improved cook stove project. The project aroused a keen interest among the participants for the positive effects it will bring to the rural communities. The stakeholders were appraised that CQC is well experienced in undertaking cookstove program in several African countries. A great expectation was generated in relation to the start of the activities for the project and achieve the expected objectives. The stakeholder also agreed that the stove project will increase the standard of living for the rural households as it will bring great impact to the communities. It was considered timely and decisive, and the initiative of C-Quest in launching this project in Philippines was praised.

The further details of local stakeholders meeting such as, newspaper notice, email invitation, Q&A , photographs etc., are provided in Appendix 1.

Further, PP will ensure that regular monitoring/field visits will be conducted by the field staff, partners or project managers to take feedback from the local stakeholders. As a part of on-going communication with local stakeholders, end users were informed about the grievance register which will be maintained at the office locations of the PP/Common place. Thus, it was explained during the meeting that end users/stakeholders are free to voice their grievances regarding the project activity anytime in the grievance register. All the relevant concerns received during operation of project activity will be addressed according to its merit.

2.3 Environmental Impact

No negative environmental impacts have been identified from the project and environmental impact assessment (EIA) is not required for the project.

2.4 Public Comments

The public comment period for the project was from 19-August-2022 to 18-September-2022. No public comments were received for the grouped project.

2.5 AFOLU-Specific Safeguards

This section is not applicable as the project is a non-AFOLU project.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, Version 1.1¹².

Tool 30- “Calculation of the fraction of non-renewable biomass”; version 04.0¹³

Standard – Sampling and Surveys for CDM project activities and programmes of activities, version 09¹⁴

3.2 Applicability of Methodology

Applicability criterion	How the project complies
Project activities shall be implemented in domestic premises or in community-based kitchen	The proposed project involves deployment of ICS only in households.
The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer’s specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	SSM S32-X stove and the Baldwin stove distributed under this project are single pot wood cookstoves that have efficiency of 51.5% and 45.0% as per the manufacturer’s specifications. Other suitable ICS models/ locally developed ICS may be added based on requirement of end users in future project activity instances meeting the applicability criteria.
Both ‘Projects’ and ‘Large Projects’ can use this methodology.	The proposed project is expected to generate 2,590,361 tonnes CO _{2e} per year which is greater than 300,000 tonnes CO _{2e} per year and thus qualifies as a “large project” as per section 3.10.1 of the VCS Standard version 4.4
Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to	Non-renewable biomass has been used since 31-December-1989 in Philippines as demonstrated at below.

¹² <https://verra.org/wp-content/uploads/2021/07/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.pdf>

¹³ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf>

¹⁴ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20210531160756223/Meth_Stan05.pdf

published literature, official reports or statistics;	
For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips) it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and (c) Energy use for renewable biomass processing (e.g., shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded	Not applicable. The ICS is introduced as energy efficiency measure to replace baseline stoves and reduce the use of non-renewable woody biomass for combustion.
The project activity shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g. programme logo).	The project proponent will ensure that every cookstove that will be distributed within the project area will have a unique ID comprising of a combination of customer name, geographical location as well as a serial number to identify the ICS end users of this project activity. In addition, the project will be cross checked against other CDM project activity operating in the country using the UNFCCC, the Gold Standard, and other relevant voluntary carbon schemes to ensure that the ICS is not included in any other CDM project activity or voluntary project activity.
The project activity shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	An undertaking will be taken by all the beneficiaries, i.e., the end-users indicating that the ownership of the current project solely lies with C-Quest Capital Stoves Asia Limited. To further ensure that the ownership rights of the project are reserved with CQC, all the stakeholders relevant, i.e., manufacturers/wholesale providers will provide their which will clearly state that the project proponent or an entity authorized by it shall be the sole owner of the VCUs arising from the project.

Applicability of TOOL30 Version 04

Applicability criterion	How the project complies
3. (a) DNAs to submit region- or country-specific default f_{NRB} values, following the procedures for development, revision, clarification, and update of standardized baselines (SB procedures); or	Not applicable
3. (b) project participants to calculate project- or PoA-specific f_{NRB} values.	PP engaged a third-party competent agency (C4 EcoSolutions) to calculate the country specific f_{NRB} based on Tool 30, version 04. f_{NRB} calculation sheet submitted to VVB.

Evidence that the non-renewable biomass has been in use since 1989

Non-renewable biomass has been in use since 31-December-1989 as evidenced by various FAO statistical data. The Global Forest Resources Assessment 2020¹⁵ (FAO) indicates that forest areas decline yearly. Between years 1990 and 2000, Philippines lost on average of 46,955 hectares of forest per year. This amounts to an average annual deforestation rate of 0.60%. Between years 2000 and 2010, the rate of forest loss is 6.4% or 0.64% per annum. In total, between 1990 and 2010, Philippines lost 12.07% of its forest cover, or around 939,090 hectares.

Decline in forest area between 1990 – 2010

Forest Resource Assessment (FRA) categories	Area (1000 hectares)		
	1990	2000	2010
Forest	7,778.81	7,309.26	6,839.72
Other wooded land	8,327.92	7,759.70	7,191.47
Other land	13,710.27	14,748.04	15,785.81
TOTAL	29,817.00	29,817.00	29,817.00

Thus, it can be concluded that losses have been reported in forest areas from before 1989 thereby hinting that fuelwood used across Philippines comes from non-renewable sources.

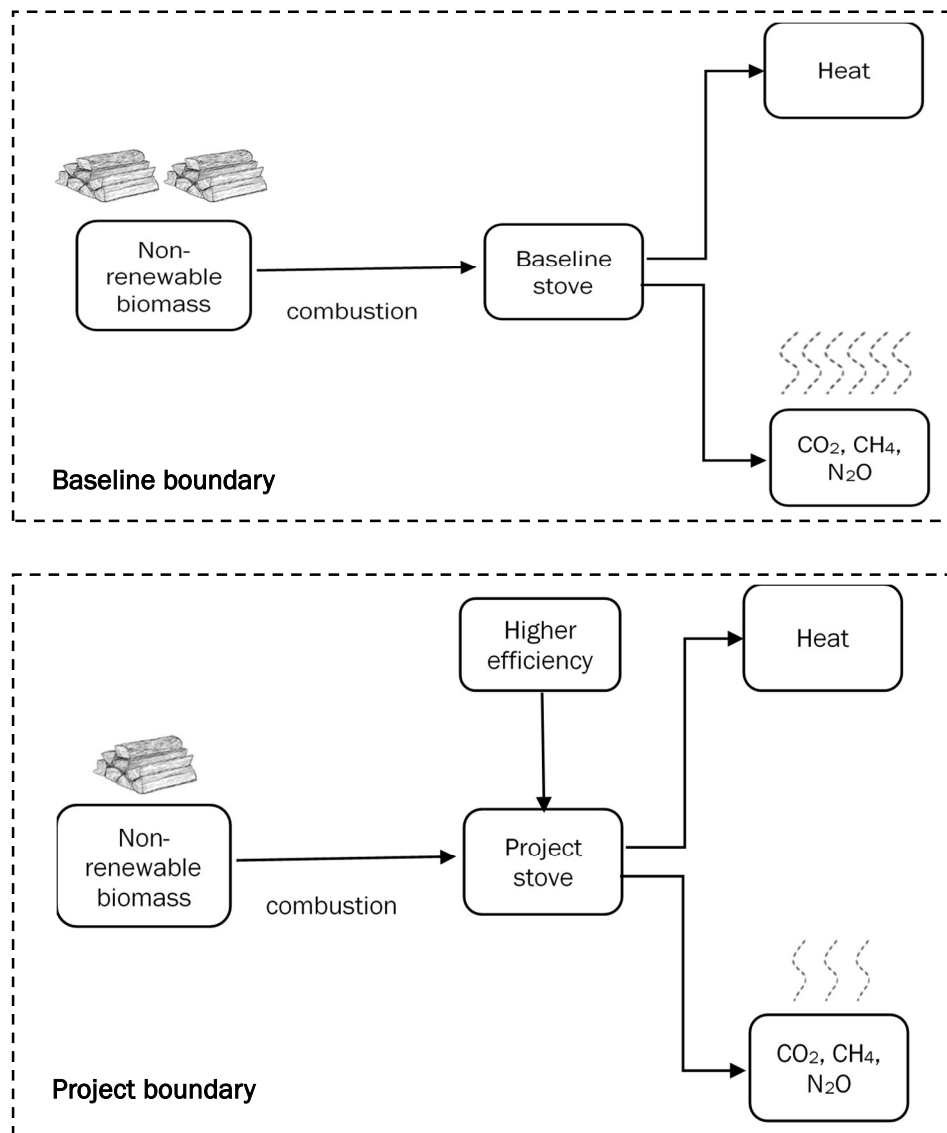
3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	Emission from use of non-renewable	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source

¹⁵ FAO, Global Forest Resources Assessment 2020, Country Report, Philippines, page 16
<https://www.fao.org/3/cb0046en/cb0046en.pdf>

	biomass/Fossil fuel	Other	No	No other source identified
Project	Emission from use of non-renewable biomass	CO ₂	Yes	Major source
		CH ₄	Yes	Major source
		N ₂ O	Yes	Major source
		Other	No	No other source identified

A representation of the baseline boundary and project boundary are given as below.



The project boundary for the grouped project consists of the physical, geographical locations of the distributed ICS limited to within the Republic of Philippines”.

3.4 Baseline Scenario

The baseline scenario is the continued use of three stone fired cookstoves or a conventional device with no improved combustion air supply or flue gas ventilation (that is without a grate or chimney) by the target population by burning non-renewable woody biomass to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

As per the applied methodology VMR0006 Version 1.1¹⁶, the efficiency of the baseline stove (three-stone cookstove) considered is 10%.



Prior to the launch of the project, PP carried out a sample baseline assessment in July 2022 to understand the baseline practices and needs of the communities. This assessment evidences that there is a clear need for an ICS program in the country. Details of the same were shared with VVB.

In addition, as per the ADB report “Increasing Access to Clean Cooking in Philippines” access to clean cooking in the Philippines was only 46% by 2018¹⁷, or an increase of only 8%¹⁸ from 2010 to 2018, leaving around 54% or 57.6 million of the population relying on traditional cookstoves utilizing fuelwood or charcoal as cooking fuel, without access to clean cooking.

As per the literature review of Philippines DHS 2022¹⁹, the percentage population using solid fuels for cooking is 41.4%, and 57.7% of the population uses clean fuels and technologies and the prevalence of traditional cookstoves exclusively using wood is about 32.6% of the total cooking population.

Also about 18%²⁰ of the households in Philippines are using Three-stone stove/open fire stoves for cooking that equals to about 5 million households (Philippines population²¹ is 113.88 million and

¹⁶ <https://verra.org/wp-content/uploads/2021/07/VMR0006-Methodology-for-Installation-of-High-Efficiency-Firewood-Cookstoves-v1.1.1.pdf> (Section 9.2 , parameter table of baseline stove efficiency)

¹⁷ <https://sdghelpdesk.unescap.org/sites/default/files/2021-03/increasing-access-clean-cooking-philippines.pdf> (pg.13/72)

¹⁸ <https://sdghelpdesk.unescap.org/sites/default/files/2021-03/increasing-access-clean-cooking-philippines.pdf> (pg.16 /72)

¹⁹ <https://dhsprogram.com/pubs/pdf/FR381/FR381.pdf> (pg. 57/617)

²⁰ <https://dhsprogram.com/pubs/pdf/FR381/FR381.pdf> (pg. 57/617)

²¹ <https://data.worldbank.org/indicator/SP.POP.TOTL?locations=PH> (for Year 2021)

average HH size of 4.2²² gives total households of 27.78 million of which 18% is 5.0 million households) are practicing three stone fire/open fire method for cooking. This clearly demonstrates the need for a large scale clean cookstove/ICS program in the Philippines. The target households under this grouped project is 500,000, which is only 10% of the total HHs practising three stone fire /open fire method for cooking.

Baseline scenario can also be verified from the end user agreement signed by the registered stove owner at the time of distribution of the project device (ICS), which confirms that, registered user voluntarily adopting the project stove and was using the three stone fire or conventional device (without a grate or a chimney) before adopting the project stove.

3.5 Additionality

The methodology uses an activity method for the demonstration of additionality.

Activity Method

Step 1: Regulatory Surplus

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework.

Households may only participate voluntarily in this project. It is hereby confirmed that the proposed project is a voluntary coordinated action by CQC.

Step 2: Positive List

As per Section 3.2, the project meets the applicability conditions of the methodology which represent the positive list.

The project installs the ICS at zero cost to the household and has no other source of revenue other than the sale of GHG credits.

The project is not implemented as part of government schemes or supported by multilateral funds.

Conclusion: As the project fulfills the conditions above, it is deemed additional.

Step 3: Project Method

This step is not applicable under project activity because the stoves are provided at zero cost to the end-users.

²² <https://dhsprogram.com/pubs/pdf/FR381/FR381.pdf> (pg. 49/617)

3.6 Methodology Deviations

The project did not apply any methodology deviations.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The applied methodology does not account for baseline emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4

4.2 Project Emissions

The methodology does not account for project emissions separately, but instead quantifies net emission reductions achieved by the project. Please refer to Section 4.4.

4.3 Leakage

Leakage considered as default 0.95 in accordance with the applied methodology VMR0006 V1.1 item 8.3.

4.4 Net GHG Emission Reductions and Removals

The improved cookstove is introduced as energy efficiency measure in the project, therefore equations 1 and 2 of the methodology will be applied to calculate the net GHG emission reductions.

$$ER_y = \sum_i \sum_j ER_{y,i,j} \quad \text{Equation (1)}$$

Where:

i	=	Indices for the situation where more than one type/model of improved cookstove is introduced to replace three-stone fire
J	=	Indices for the situation where there is more than one batch of improved cookstove of type i
ER_y	=	Emission reductions during year y in t CO ₂ e
$ER_{y,i,j}$	=	Emission reductions by improved cookstove of type i and batch j during year y in t CO ₂ e

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95 \quad \text{Equation (2)}$$

Where:

$B_{y,savings,i,j}$	=	Quantity of woody biomass that is saved in tonnes per improved cookstove of type i and batch j during year y
$f_{NRB,y}$	=	Fraction of woody biomass that can be established as non-renewable biomass (fNRB) ²³
$NCV_{wood\ fuel}$	=	Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default for wood fuel, 0.0156 TJ/tonne) ²⁴
EF_{wf,CO_2}	=	CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 112 tCO ₂ /TJ) ²⁵
$EF_{wf,non\ CO_2}$	=	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel, 26.23 tCO ₂ /TJ) ²⁶
$N_{y,i,j}$	=	Number of improved cookstoves of type i and batch j operating during year y
0.95	=	Discount factor to account for leakage

The quantify of woody biomass saved due to implementation of improved cookstoves to be estimated using equation below:

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right) \quad \text{Equation (3)}$$

where

η_{old}	=	Efficiency of baseline cookstove
$\eta_{new,y,i,j}$	=	Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y Alternatively, efficiency may be determined using Equation 4.
$B_{y=1,new,i,j,survey}$	=	Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i and batch j , determined in the first year of the implementation of the project through a sample survey.

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94 \quad \text{Equation (4)}$$

²³ C4 Eco Solutions (Pty) Ltd was appointed as third party to study and derive the fNRB value for Philippines and the report submitted to VVB.

²⁴ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction

²⁵ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

²⁶ 2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion

where

- η_p = Efficiency of project stove (fraction) at the start of project activity.
- $(DF_n)^{y-1}$ = Discount factor to account for efficiency loss of project cookstove per year of operation (fraction). This value may be based on actual monitoring or based on manufacturer's declaration on expected loss in efficiency or through publicly available literature on relevant industry standards. Alternatively default value of 0.99 efficiency loss per year can be considered.
- 0.94 = Adjustment factor to account for uncertainty related to project cookstove efficiency test.

To calculate f_{NRB} the equations as follows:

$$f_{NRB} = \frac{NRB}{NRB + RB} \quad \text{Equation (5)}$$

$$NRB = H - RB \quad \text{Equation (6)}$$

$$H = HW \times N + C + NE \quad \text{Equation (7)}$$

$$RB = \sum (MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other,i})) \quad \text{Equation (8)}$$

Where,

- NRB = non-renewable woody biomass
- RB = renewable woody biomass production
- H = Total annual wood consumption (energy/non energy) in the country
- HW = Average HH wood fuel (including charcoal) consumption²⁷
- N = Number of HHs consuming wood fuel in the region/project area.
- CE = Commercial woody biomass consumption for energy application.²⁸
- NE = Commercial woody biomass consumption for non- energy application

²⁷ Approved standardized baseline; Published literature/research paper –Government data, organizations of repute (FAO, UN etc.);

²⁸ Energy Statistics Database -UN Statistics Division; Existing studies; Government data; Survey.

- MAI = Mean annual increment in forest/other wooded land²⁹
- F = Extent of forest/other wooded land³⁰
- P = Extent of protected area (including non-accessible area) within forest/other wooded land.³¹

Fraction of non-renewable biomass (f_{NRB}) calculation for Philippines:

Parameter	Value	Source
Total Woody biomass consumption (t/yr) (H)	17,301,664	The World Bank ³² , FAO Statistics ³³ (in accordance with section 3.1 para 14 of the TOOL30, v04.0).
Renewable biomass (t/yr) (RB)	7,178,624	UN Data, 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Local study (in accordance with section 3.2 para 19 of the TOOL30, v4.0)
Non-renewable biomass (t/yr) (NRB)	10,123,040	Calculated (Equation 2 of applied CDM tool 30, v04.0)
Fraction of non-renewable biomass(f_{NRB})	0.59	Calculated ³⁴ (Equation 1 of applied CDM Tool 30, v04.0)

The resulting non-renewable biomass (NRB) was compared to the top-down product of average above-ground biomass (219.77 t/ha) and the most recent annual deforestation data (34,890 ha/yr). NRB, as calculated according to the latest CDM Tool 30 v4.0 2022, was found to be 10,123,040 t/yr, approximately the same as the cross-check results based on deforestation (7,667,775 t/yr). The cross-check estimate of biomass loss from deforestation could underestimate the extent of non-renewable biomass as it only considers the average biomass stocks from completely deforested areas. Unsustainable wood harvesting resulting in forest degradation and substantial reductions in biomass stocks, but not complete deforestation is not considered in the cross-check. The NRB estimate is, therefore, considered applicable for this study, given that the guidance provided by the latest CDM Tool 30 v4.0 2022 was followed.

The fraction of non-renewable biomass is the quotient of the non-renewable (NRB) and the total biomass (Equation 5 above). The fraction of non-renewable biomass for Philippines is, therefore, estimated to be 0.59.

²⁹ IPCC Good Practice Guidelines, Table 4.9, [www.ipcc-nggip.iges.or.jp](http://www.ipcc-nggip.iges.or.jp/www/fao.org) National studies or government data or official statistics.

³⁰ Global Forest Resources Assessment by the Food and Agriculture Organization of the United Nations (FAO); Official statistics: Project specific survey data.

³¹ Global Forest Resources Assessment by the Food and Agriculture Organization of the United Nations (FAO); National studies or Government data or Official statistics

³² <https://data.worldbank.org/>

³³ <https://www.fao.org/3/cb3795m/cb3795m.pdf>

³⁴ Calculated as $f_{NRB} = NRB / (NRB + RB)$

The resulting f_{NRB} value is higher than the expected f_{NRB} values determined in peer reviewed literature such as “The carbon footprint of traditional woodfuels” by Bailis and colleagues using the WISDOM method. While direct comparisons between the WISDOM and CDM methodologies are only sometimes appropriate, given the different approaches of the methodologies, the following factors can contribute to variations in the f_{NRB} estimates between the application of the two methods:

- i) Population statistics (2021) for Philippines were used and represents a 16.8%³⁵ increase in population numbers since the 2009 data as utilised in the comparison study.
- ii) Updated 2019 FAO Forest Products statistics were used in the present study, whereas the comparison study used values prior to 2013 from either FAO Forest Products statistics, 2013 United Nations (UN) Energy Database or country specific reports depending on the availability of consumption data.
- iii) The approach used to determine the area of forest that is accessible for wood harvesting yields different estimates in the present study than the comparison study. Given that the accessible harvesting area is used to determine the RB, differences in this estimate can lead to differences in the f_{NRB} . Moreover, in the present study, recent data was used for global administrative zones, roads, protected areas and tree cover change from which accessibility is determined.
- iv) In the present study, MAIs were calculated using a weighted average based on the forest area of three categories (i.e., primary forests, above 20-year secondary forest, below 20-year secondary forest). Data from the 2019 Refinement of the 2006 IPCC Guidelines was used in combination with extrapolating the observed forest gain extents between 2000 and 2012 to a future 20-year period. As per the study by Bailis and colleagues, MAI values were derived from a combination of field observations and IPCC values, followed by a different estimation of growth rates as a percentage of standing stock. This approach often yields higher MAIs and may lead to higher estimations of RB and subsequently, lower estimations of NRB and f_{NRB} .

Moreover, the different approach to determining the MAI and non-domestic consumption, as stipulated by the latest CDM Tool 30 v4.0 2022 in comparison to the WISDOM method used in the above-mentioned peer reviewed literature study, and the differences as stated above vide item i) to iv) provides the variance in the expected f_{NRB} values.

For ex-ante calculation purpose, the assumption below is applied.

- 1) It is assumed that the 1,000,000 ICS to be distributed under SSM S32-X stove and Baldwin stove with estimated proportion of 50:50.
- 2) Annual stove loss rate is estimated at 10% for SSM S32-X stove and 10% for Baldwin stove.
- 3) $B_{y=1,new,i,j,survey}$ is assumed as 1.15 tonnes / device / year.
- 4) Fraction of non-renewable biomass (f_{NRB}) = 0.59³⁶

³⁵ The values provided in f_{NRB} Calculation excel file under Total population sheet

³⁶ The f_{NRB} value determined ex ante as per methodological CDM Tool 30, Version 04 by third party and the f_{NRB} calculations sheet provided to VVB.

Determination of number of ICS operating during year y

$$N_{y,i,j} = 500,000 \times [1 - (y-1) \times 10\%]$$

Example of calculation:

If $y = 2$,

$$N_{y,i,j} = 500,000 \times [1 - (2-1) \times 10\%]$$

$$= 500,000 \times 90\%$$

$$= 450,000$$

Hence, the number of ICS operating during year y is as below:

	SSM S32-X stove	Baldwin Stove
Year (y)	$N_{y,i,j}$	$N_{y,i,j}$
1	500,000	500,000
2	450,000	450,000
3	400,000	400,000
4	350,000	350,000
5	300,000	300,000
6	250,000	250,000
7	200,000	200,000
8	150,000	150,000
9	100,000	100,000
10	50,000	50,000

Determination of efficiency of ICS during year y

$$\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$$

Where

$$\eta_p = 51.5\%$$

$$DF_n = 0.99$$

Example of calculation:

If $y = 2$

$$\eta_{new,y,i,j} = 51.5\% \times (0.99)^{2-1} \times 0.94$$

$$= 47.93\%$$

Hence the efficiency of ICS models during year y is as below:

	SSM S32-X stove	Baldwin Stove
Year (y)	$\eta_{new,y,i,j}$	$\eta_{new,y,i,j}$
1	48.41%	42.30%
2	47.93%	41.88%
3	47.45%	41.46%
4	46.97%	41.04%
5	46.50%	40.63%
6	46.04%	40.23%
7	45.58%	39.82%
8	45.12%	39.43%
9	44.67%	39.03%
10	44.22%	38.64%

Determination of quantity of woody biomass that is saved in tonnes per ICS during year y

$$B_{y,savings,i,j} = B_{y=1,new,i,survey} \times \left(\frac{\eta_{new,y,i,j}}{\eta_{old}} - 1 \right)$$

Example of calculation for SSM-S32X stove:

If $y = 2$,

$$B_{y,savings,i,j} = 1.15 \times [(0.4793/0.1) - 1]$$

$$= 4.3439 \text{ tonnes}$$

	SSM S32-X stove	Baldwin Stove	SSM S32-X stove	Baldwin Stove
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Year (y)	$B_{y=1,new,i,survey}$	η_{old}	$\eta_{new,y,i,j}$	$\eta_{new,y,i,j}$	$B_{y,savings,i,j}$	$B_{y,savings,i,j}$
1	1.15	0.1	48.41%	42.30%	4.3994	3.6995
2	1.15	0.1	47.93%	41.88%	4.3439	3.6511
3	1.15	0.1	47.45%	41.46%	4.2890	3.6031
4	1.15	0.1	46.97%	41.04%	4.2347	3.5556
5	1.15	0.1	46.50%	40.63%	4.1809	3.5086
6	1.15	0.1	46.04%	40.23%	4.1276	3.4621
7	1.15	0.1	45.58%	39.82%	4.0749	3.4160
8	1.15	0.1	45.12%	39.43%	4.0227	3.3704
9	1.15	0.1	44.67%	39.03%	3.9710	3.3252
10	1.15	0.1	44.22%	38.64%	3.9198	3.2805

Determination of emission reductions by ICS during year y

$$ER_{y,i,j} = B_{y,savings,i,j} \times NCV_{wood\ fuel} \times f_{NRB,y} \times (EF_{wf,CO_2} + EF_{wf,non\ CO_2}) \times N_{y,i,j} \times 0.95$$

Where

$$NCV_{wood\ fuel} = 0.0156 \text{ TJ/tonne}$$

$$f_{NRB,y} = 0.59$$

$$EF_{wf,CO_2} + EF_{wf,non\ CO_2} = 112 + 26.23 = 138.23 \text{ tCO}_2/\text{TJ}$$

Example of calculation:

SSM S32-X stove

If y=2,

$$ER_{y,i,j} = 4.3439 \times 0.0156 \times 0.59 \times 138.23 \times 450,000 \times 0.95$$

$$= 2,362,635.56 \text{ tCO}_2$$

Year	$B_{y,savings,i,j}$	$NCV_{wood\ fuel}$	$f_{NRB,y}$	$EF_{wf,CO_2} + EF_{wf,non\ CO_2}$	$N_{y,i,j}$	$ER_{y,i,j}$
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1	4.3994	0.0156	0.59	138.23	500,000	2,658,659.00
2	4.3439	0.0156	0.59	138.23	450,000	2,362,635.56
3	4.2890	0.0156	0.59	138.23	400,000	2,073,581.86
4	4.2347	0.0156	0.59	138.23	350,000	1,791,395.04
5	4.1809	0.0156	0.59	138.23	300,000	1,515,973.57
6	4.1276	0.0156	0.59	138.23	250,000	1,247,217.30
7	4.0749	0.0156	0.59	138.23	200,000	985,027.39
8	4.0227	0.0156	0.59	138.23	150,000	729,306.30
9	3.9710	0.0156	0.59	138.23	100,000	479,957.80
10	3.9198	0.0156	0.59	138.23	50,000	236,886.93

Example of calculation:

Baldwin stove

If $y=2$,

$$ER_{y,i,j} = 3.6511 \times 0.0156 \times 0.59 \times 138.23 \times 450,000 \times 0.95$$

$$= 1,985,812.70 \text{ tCO}_2$$

Year	$B_{y,savings,i,j}$	$NCV_{wood\ fuel}$	$f_{NRB,y}$	$EF_{wf,CO2}$ + $EF_{wf,non\ CO2}$	$N_{y,i,j}$	$ER_{y,i,j}$
1	3.6995	0.0156	0.59	138.23	500,000	2,235,737.72
2	3.6511	0.0156	0.59	138.23	450,000	1,985,812.70
3	3.6031	0.0156	0.59	138.23	400,000	1,741,977.74
4	3.5556	0.0156	0.59	138.23	350,000	1,504,142.97
5	3.5086	0.0156	0.59	138.23	300,000	1,272,219.67
6	3.4621	0.0156	0.59	138.23	250,000	1,046,120.34
7	3.4160	0.0156	0.59	138.23	200,000	825,758.59
8	3.3704	0.0156	0.59	138.23	150,000	611,049.22
9	3.3252	0.0156	0.59	138.23	100,000	401,908.13

10	3.9759	0.0156	0.59	138.23	50,000	198,252.34
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	SSM S32-X stove	Baldwin stove	Total (round down)
Year	$ER_{y,i,j}$	$ER_{y,i,j}$	tCO ₂
1	2,658,659.00	2,235,737.72	4,894,396
2	2,362,635.56	1,985,812.70	4,348,448
3	2,073,581.86	1,741,977.74	3,815,559
4	1,791,395.04	1,504,142.97	3,295,538
5	1,515,973.57	1,272,219.67	2,788,193
6	1,247,217.30	1,046,120.34	2,293,337
7	985,027.39	825,758.59	1,810,785
8	729,306.30	611,049.22	1,340,355
9	479,957.80	401,908.13	881,865
10	236,886.93	198,252.34	435,139

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
15-February 2023 to 14-February 2024	-	-	-	4,894,396
15-February-2024 to 14-February-2025	-	-	-	4,348,448
15-February-2025 to 14-February-2026	-	-	-	3,815,559
15-February-2026 to 14-February-2027	-	-	-	3,295,538
15-February-2027 to 14-February-2028	-	-	-	2,788,193
15-February-2028 to 14-February-2029	-	-	-	2,293,337
15-February-2029 to 14-February-2030	-	-	-	1,810,785

15-Febrary-2030 to 14-Febrary-2031	-	-	-	1,340,355
15-Febrary-2031 to 14-Febrary-2032	-	-	-	881,865
15-February-2032 to 14-Febrary-2033	-	-	-	435,139
Total	-	-	-	25,903,615

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction
Description	Fraction of woody biomass saved by the project activity during year y that can be established as non-renewable biomass
Source of data	Survey report (IPCC & FAO reports)
Value applied	0.59
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante. C4 EcoSolutions (Pty) Ltd was appointed as third party to study and derive the f_{NRB} value for Philippines.
Purpose of Data	Calculation of emission reductions
Comments	The f_{NRB} calculations sheet provided to VVB .

Data / Parameter	$NCV_{wood\ fuel}$
Data unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted or reduced
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 1 Introduction

Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	EF_{wf,CO_2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	$EF_{wf,non\ CO_2}$
Data unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor for the use of wood fuel in baseline scenario
Source of data	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy, Chapter 2 Stationary Combustion
Value applied	26.23
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value

Purpose of Data	Calculation of emission reductions
Comments	-

Data / Parameter	η_p
Data unit	Fraction
Description	Efficiency of project stove at the start of project activity.
Source of data	Manufacturer's specification
Value applied	SSM S32-X stove: 51.5% Baldwin stove: 45.0%
Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of Data	Calculation of $\eta_{new,y,i,j}$
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	$N_{y,i,j}$
Data unit	Number
Description	Number of project devices of type I and batch j operating during year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and programme of activities.
Frequency of monitoring/recording	At least once every two years

Value applied	For ex-ante emission reduction calculation, it is assumed that the project will distribute up to 500,000 ICS with proportion of 50:50 between SSM S32-X stove and Baldwin stove.
Monitoring equipment	Monitoring survey
QA/QC procedures to be applied	Sampling standard “sampling and surveys for CDM project activities and programme of activities” version 9 will be used for determining the sample size to achieve 90/10 confidence precision.
Purpose of data	Calculation of emission reductions
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value
Comments	No Comments

Data / Parameter	η_{old}
Data unit	Fraction
Description	Efficiency of baseline cookstove
Source of data	Methodological default value
Description of measurement methods and procedures to be applied	As per the applied methodology default value of 0.1 shall be used if baseline device is a three-stone fire using firewood (not charcoal), or a conventional device with no improved combustion air supply or flue gas ventilation, that is without a grate or a chimney.
Frequency of monitoring/recording	Fixed for each individual household at the time of project implementation.
Value applied	0.1
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Calculation of emission reductions
Calculation method	none
Comments	As per the methodology, this value has been fixed for each household that receives project ICS, during the project crediting period.

Data / Parameter	$\eta_{new,y,i,j}$																														
Data unit	Fraction																														
Description	Efficiency of the improved cookstove type i and batch j determined through water boiling test (WBT) during year y																														
Source of data	Calculation																														
Description of measurement methods and procedures to be applied	To adopt Option V given in the methodology: “Efficiency of the improved cookstoves to be estimated using equation 4 above where loss in efficiency per year is calculated, and therefore this parameter does not need to be monitored”																														
Frequency of monitoring/recording	Annually																														
Value applied	For ex-ante calculation, the value below is applied. SSM S32-X <table border="1"> <thead> <tr> <th>Year (y)</th><th>$\eta_{new,y,i,j}$</th></tr> </thead> <tbody> <tr><td>1</td><td>48.41%</td></tr> <tr><td>2</td><td>47.93%</td></tr> <tr><td>3</td><td>47.45%</td></tr> <tr><td>4</td><td>46.97%</td></tr> <tr><td>5</td><td>46.50%</td></tr> <tr><td>6</td><td>46.04%</td></tr> <tr><td>7</td><td>45.58%</td></tr> <tr><td>8</td><td>45.12%</td></tr> <tr><td>9</td><td>44.67%</td></tr> <tr><td>10</td><td>44.22%</td></tr> </tbody> </table> Baldwin stove <table border="1"> <thead> <tr> <th>Year (y)</th><th>$\eta_{new,y,i,j}$</th></tr> </thead> <tbody> <tr><td>1</td><td>42.30%</td></tr> <tr><td>2</td><td>41.88%</td></tr> <tr><td>3</td><td>41.46%</td></tr> </tbody> </table>	Year (y)	$\eta_{new,y,i,j}$	1	48.41%	2	47.93%	3	47.45%	4	46.97%	5	46.50%	6	46.04%	7	45.58%	8	45.12%	9	44.67%	10	44.22%	Year (y)	$\eta_{new,y,i,j}$	1	42.30%	2	41.88%	3	41.46%
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5	40.63%														
6	40.23%														
7	39.82%														
8	39.43%														
9	39.03%														
10	38.64%														
Monitoring equipment	Calculated value (No Monitoring equipment used)														
QA/QC procedures to be applied	This parameter will be calculated using equation 5 of the applied methodology.														
Purpose of data	Calculation of emission reductions														
Calculation method	Calculation to be performed using equation below: $\eta_{new,y,i,j} = \eta_p \times (DF_n)^{y-1} \times 0.94$														
Comments	Fixed Ex-ante														

Data / Parameter	$B_{y=1,new,i,j,survey}$
Data unit	Tonnes
Description	Annual quantity of woody biomass used by improved cookstoves in tonnes per device of type i and batch j, determined in the first year of the implementation of the project through a sample survey
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	Minimum sample size of each type i and batch j should be in line with the latest version of Standard for sampling and surveys for CDM project activities and programme of activities or guidelines provided in methodology Section 8.4 option (b). Determined in the first year of the introduction of the devices (e.g. during the first year of the crediting period, $y=1$) through measurement campaigns at representative households and/or sample survey. Sample surveys to estimate this parameter, that are solely based on questionnaires or interviews (i.e. that do not implement

	measurement campaigns) may only be used if the following conditions are satisfied. (a) Baseline cookstoves have been completely decommissioned and only improved cookstoves are exclusively used in the project households; (b) If multiple devices are used in the project, it is possible from the results of the survey questions to clearly differentiate the quantity of firewood being used by each device. In other words, if more than one device, or another device that consumes firewood, are in use in project households, then the sample survey needs to distinguish the quantity of firewood used by the project device and the other devices that use firewood.
Frequency of monitoring/recording	Determined in the first year of project implementation
Value applied	For ex-ante calculation, the value is assumed as 3.138 kg/device/day or equal to 1.15 tonnes/device/year. (based on the survey results of the similar project implemented in Vietnam (a country in Southeast Asia) with VCS ID 2523)
Monitoring equipment	Weighing Scales to be used for measuring quantity of fuel wood
QA/QC procedures to be applied	Calibration of weighing scales to be used for measuring the fuel wood will get done before start using on site
Purpose of data	Calculation of emission reductions
Calculation method	Not Applicable.
Comments	If any of the above conditions (a) and (b) are not met the PP will undertake measurement for determining the parameter $B_{y=1,new,l,j,survey}$

Data / Parameter	Life Span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the project stove efficiency
Source of data	Manufacturer's specification
Description of measurement methods and procedures to be applied	Technical specification as declared by the manufacturer and wherever possible supported by third party report.

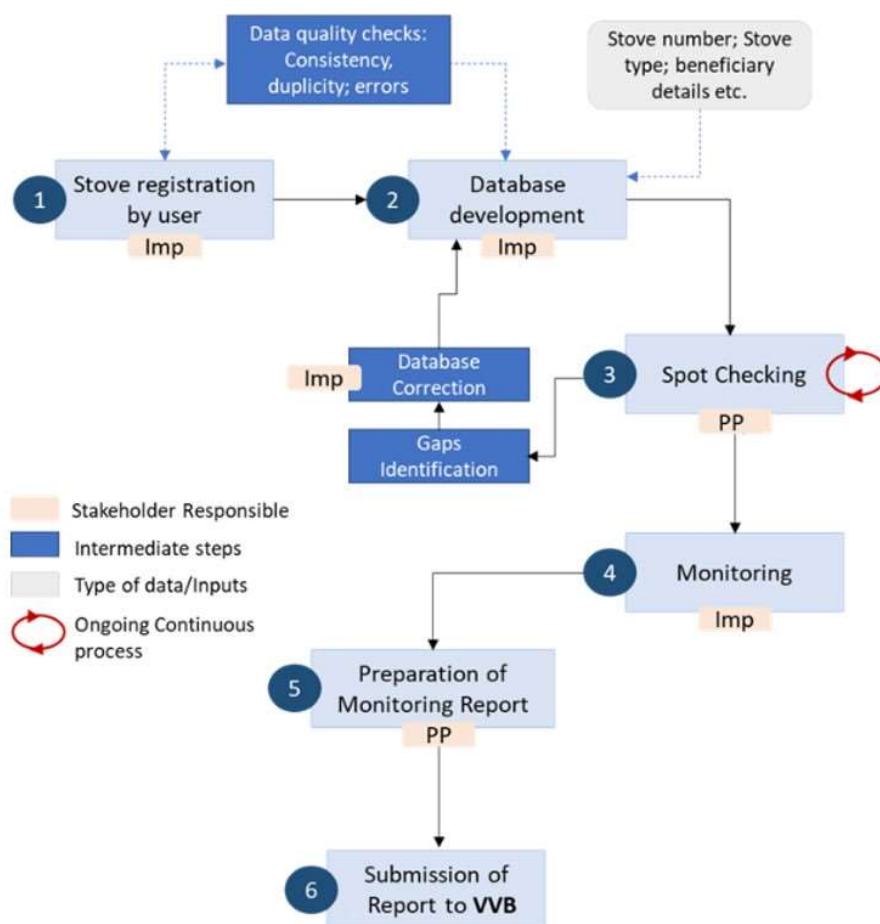
Frequency of monitoring/recording	Once at the time of project stove installation
Value applied	SSM S32-X stove: 10 years Baldwin stove: 10 years
Monitoring equipment	No equipment is required to monitor this parameter
QA/QC procedures to be applied	This parameter will be referred from the Manufacturer's specification
Purpose of data	Calculation of emission reductions
Calculation method	No calculation used for this parameter
Comments	If the life span of project devices is less than the crediting period, it shall be demonstrated that the devices shall be replaced after the life span has ended. In such cases, if it cannot be demonstrated that the project devices will be replaced with new devices, no emission reductions can be claimed beyond the life span of the project devices.

5.3 Monitoring Plan

The Monitoring Plan applied in this project involves several key elements that ensure that the PP and/or project Implementer have high-quality, unbiased, and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating Verified Carbon Units (VCUs) following VCS methodology (VMR0006 version 1.1) on the basis of the amount of non-renewable biomass saved by the ICS in the project activity. The key elements are the following:

- Data collection procedures
- Distribution and Monitoring Database
- Spot Checking of ICS (ongoing)
- Sample Plan for the Monitoring Survey
- Data Quality, Consistency and Duplication Checks
- Monitoring Reporting

The below flow-chart illustrates the roles and responsibilities of the parties during the implementation of the monitoring plan for the project. In the below flowchart, the project implementer is abbreviated to "Imp" and can be the PP or another party authorized by the PP.



Roles and Responsibilities of stakeholders during the implementation of the monitoring plan

Below is the description of the above steps on the flow-chart.

1. **Implementor: User registers stove:** Project implementer will collect/receive the necessary information requested in the Registration process from the user. Means of collecting this information may be through a physical registration card filled by project Implementor staff, retailers, end-users, or partner organization's staff, or using ICTs or SMS. Project Implementers' staff shall double check the accuracy of the information provided, and request for field staff additional clarifications if needed.
2. **Implementor: Data logged into database:** Project implementer's trained staff will input the data in the database either manually (if data collected from physical registration card) or this will be automatically input if data was collected using ICTs or SMS. Project implementer staff shall double check the information included on the database and check for duplications. Any duplicate information shall be investigated, and errors corrected or excluded from the database if it is a true duplicate entry.
3. **Project Proponent: Spot- checking (ongoing):** Project Proponent has a procedure for internal auditing called spot-checking, where field staff randomly selects households to ensure proper functioning of the project stoves throughout the project lifetime and to ensure that the project stoves continue to work as efficient as specified by the manufacturer. Field staff of project proponent randomly select units included in the database and visit or contact the stove owners

to cross-check the information on the database with the factual evidence in the field. Any inconsistencies found (e.g., change in the address of a user) get updated on the database, and in case, ICS are found to be no longer in use, they get clearly marked as such and is excluded from emission reductions calculations.

During the spot check, if any of the project stoves was found to be in poor working condition like damaged or in unusable condition, the stove owner may repair the stove for minor issues. If the stove owners are unsuccessful in repairing or solving the issues on their own, the owner may reach out to CQC & IP through helpline, field personnel or the village headmen for further assistance. Respective channel representative, either headman, field staff, IP, helpline executive or authorized personnel informs CQC or their officials for appropriate action and to resolve the grievance raised by the beneficiary. CQC or their representatives will keep a record of such incidents and ensure the resolution is achieved to the satisfaction of the end users.

4. **Implementor: Monitoring:** Project implementer will follow the requirements as per PD to collect the necessary information for a monitoring report. The project proponent will assess all monitoring data and produce a monitoring report for the VVB to verify corresponding to the preceding monitoring period. This report will present the data relating to the emission reductions generated by those batches during the monitoring period.
5. **Project Proponent: Preparation of monitoring report:** the project implementers or the PP will prepare the final monitoring report to be provided to the VVB for verification of emission reductions. A copy of the monitoring report will remain with the Project Proponent.

The PP will coordinate and manage each project Implementer and assist them in implementing each element of the monitoring plan. The monitoring plan shall be elaborated in accordance with the Sampling Plan below. Also, regular audits conducted throughout the project lifetime ensures proper functioning of stoves and determination of the lifespan of the same.

(a) Sampling design

The sampling plan design in line with the requirements of the *Standard for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 09.0*.

Due to the large number of ICS envisioned to be distributed under the project, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling will be undertaken.

Samples will be drawn from ICS registered in the project database administered by the project proponent. A detailed explanation of the database with minimum information for each project device is provided in Section 1.11 of project description.

1. Objectives and Reliability Requirements

The objective is to obtain an unbiased and reliable estimate of the proportion or mean value of the following key variables over the course of the crediting period and with 90/10 confidence/precision (as per applied methodology VMRO006; version 1.1).

Monitored Parameters:

The project involves introduction of improved cook stoves as energy efficiency measure:

No.	Monitoring parameters	Sampling parameters	Parameter type	Monitoring frequency
1	$N_{y,i,j}$ Number of project devices of type i and batch j operating during year y	Proportion of ICS still in operation	Proportion	Annual/Biennially
2	$B_{y=1,new,i,j,survey}$ Quantity of woody biomass used by project devices in tonnes per device of type i and batch j	Daily consumption of woody biomass per ICS	Mean value	Determined in the first year of project implementation

II. Target Population

The target population will be the complete set of appliances (ICS) deployed under the project.

III. Sampling Method

The sampling method for the monitored parameters $N_{y,i,j}$ and $B_{y=1,new,i,j,survey}$ is Simple Random Sampling and samples will be randomly selected from the primary sampling units as illustrated above. To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the project proponent.

$N_{y,i,j}$ Visual inspection of the premises to see if ICS is operational and in use. Interview with end user if required to verify that ICS is still in use (Yes/No)

$B_{y=1,new,i,j,survey}$ Interview with end user and estimate the daily consumption of woody biomass of ICS (Daily consumption of woody biomass)

IV. Sample Size

In order to calculate the required sample size estimates, values for the proportions, mean values, and standard deviations are required. As per *Guideline for Sampling and surveys for CDM project activities and programmes of activities, version 04.0*, there are different ways available to obtain the estimates of the parameter of interest:

- (a) Refer to the result of previous studies and use these results;
- (b) In a situation where information from previous studies is not available, a preliminary sample as a pilot could be conducted and use that sample is used to provide the estimates;
- (c) Use best guesses based on the researcher's own experiences.

(i) Proportion parameter

To estimate the sample size for proportion parameters, the following equation³⁷ is used:

$$n \geq \frac{1.645^2 N \times p(1-p)}{(N-1) \times 0.1^2 \times p^2 + 1.645^2 \times p(1-p)}$$

where

- n = Sample size
- N = Population size (Total number of households or ICS)
- p = Expected proportion³⁸
- 1.645 = Represents the 90% confidence required
(In the case of 95% confidence, 1.96 shall be used)
- 0.1 = Represents the 10% relative precision

The sample size calculated above assumes 100% response, and therefore needs to be scaled up where non-response is likely to occur as per actual sampling surveys.

(ii) Mean value parameter

The minimum sample size for 'mean value' parameter (quantity of woody biomass used in the project devices in the first year of installation) required to be determined through sampling would be calculated using the equation for simple random sampling as per the Guideline: Sampling and surveys for CDM project activities and programmes of activities Version 04.0, as follows:

To estimate the sample size for mean value parameter, the following equation³⁹ is used:

³⁷ Equation 1 of Appendix 1, *Guidelines for Sampling and Surveys for CDM Project Activities and Programme of Activities (Version 04.0)*

³⁸ Using existing data or a small pilot survey, the expected proportion can be estimated for purposes of determining the minimum sample size.

³⁹ Equation 18 of Appendix 1, *Guidelines for Sampling and Surveys in CDM Project Activities and Programme of Activities (version 04.0)*

$$n \geq \frac{1.645^2 N V}{(N - 1) \times 0.1^2 + 1.645^2 V}$$

Where:

$$V = \left(\frac{SD}{mean} \right)^2$$

n = Sample size

N = Population size (Total number of households or ICS)

$mean$ = Expected mean

SD = Expected standard deviation

1.645 = Represents the 90% confidence required

(In the case of 95% confidence, 1.96 shall be used)

0.1 = Represents the 10% relative precision

If the resulting sample size based on the above equation is smaller than 30 ICS, then as the parameter of interest is a numeric mean value (i.e. not a proportion or percentage) the Student's t-distribution shall be used.

The sample size for mean value parameter is referred to the equation below⁴⁰:

$$n = \left(\frac{t_{n-1} \times SD}{0.1 \times mean} \right)^2$$

Where t_{n-1} is the value of the t-distribution for 90% confidence when the sample size is n . Since the sample size is not known yet, the first step is to use the value for 90% confidence when the sample is large, i.e. 1.645 and then redefine the calculation.

The calculation now need to repeat using t-value for 90% confidence. The process should be iterated until there is no change to the value of n .

The project proponent may choose to use the same sample to monitor more than one parameter. According to the Standard for sampling and surveys for CDM project activities and programme of activities, if there is more than one parameter to be estimated, then a sample size calculation should be done for each of them. Then either the largest number for the sample size is chosen as sampling effort with one common survey, or separate sampling efforts and surveys are undertaken for each parameter.

⁴⁰ Paragraph 102, *Guidelines for Sampling and Surveys in CDM Project Activities and Programme of Activities* (version 04.0)

For instance, the project proponent may sample separately $N_{y,i,j}$ and $B_{y=1,new,i,j,survey}$ or a combination of two parameters in the same sample. Sampling more than one parameter within the same sample (household) helps reduce travel needs for monitoring and the associated costs. At the same time this approach ensures the random selection of samples for every parameter.

The calculation above does not take into account non-response. If a level of non-response is expected within the sample then the sample size should be scaled up accordingly. (Project proponent shall determine the appropriate non-responses rate based on previous experience).

V. Sampling Frame

Overall, a particular model of ICS will have same group of end users which are household users, thus it is expected that the geographical locations do not have influence on the parameter of interest. Therefore, all monitoring parameters can be assumed to be highly homogeneous for each ICS model regardless of how the end user group and distribution/installation location is defined since the end users will always be households and geographical location is restricted to the project boundary.

The Sampling Frame for individual parameters have been defined in the following paragraphs: All the ICS distributed under this project shall be grouped together to create a Primary Sampling Unit which is homogenous. The procedure to determine the sample of households will ensure that they adequately represent the broader project population ie ICS, minimizing sampling error.

a. Sampling frame for proportion of ICS still in operation ($N_{y,j}$)

The first step is to identify the Primary Sampling Units.

Factors such as model of the ICS do not effect parameter $N_{y,j}$. Therefore Primary sampling unit in case of parameters $N_{y,j}$ is independent of model of stove.

b. Quantity of woody biomass used by project devices in tonnes per device ($B_{y=1,new,i,survey}$)

The Parameter $B_{y=1,new,i,survey}$ shall vary in accordance with its model. Hence for this parameter, the Primary Sampling Unit shall be defined as the group of ICSs of the same model. If two different ICS models being implemented in the same project – this will form two Primary Sampling Units. The value will be determined in the first year of the introduction of the devices through the monitoring survey.

Under this project up to two ICS will be installed in each household, which are classified as Project stove 1 and project stove 2. At the time of monitoring survey, field staff record the same in survey app. Therefore, firewood consumed for each project stove can be distinguished clearly. For measuring the fuel wood, calibrated weighing scales will be used to ensure data quality and accuracy.

(b) Data to be collected:

(i) Field Measurements:

The table below summarizes field measurement data requirements

Parameter	Timing (indicative)	Frequency as required by methodology	Methods to be applied	Comments on seasonal fluctuation
$N_{y,i,j}$	Monitoring will likely occur with in 12 to 24 months	At least once in two years (Biennially)	Visits to the premises, visual inspection and interview with ICS end-user.	Seasonal fluctuation is not expected to affect this parameter.
$B_{y=1,new,i,j,survey}$	Monitoring will likely occur within the first year after installation	First year of installation and fixed for crediting period.	Visits to the premises, visual inspection and interview with ICS end-user and measurement campaign.	Seasonal fluctuation is not expected to affect this parameter. However, in case the survey results show a difference, the same <u>will</u> be taken into ER estimations calculations.

(ii) Quality Assurance/Quality Control

A household level questionnaire was designed to collect information for the parameters of interest. The project proponent will apply measures to ensure the required confidence/precision for each sampled parameter is met, if precision required is not achieved by reliability check, use the lower bound or upper bound of estimates of the parameter.

The monitoring plan will explain how to properly survey households to prevent bias from interviewer. In the case a household refuses to participate, another household will be chosen at random. To reduce interviewer bias, good questionnaire design and well-tested questionnaires will be used. The project proponent or a competent third party designated by the project proponent with the proper skills will train the monitoring personnel on how to properly survey households to prevent bias from interviewer.

The PP may choose to stop monitoring a particular parameter once the required level of confidence/precision has been reached, as long as the calculated minimum number of samples has been achieved.

To identify and address outliers for the parameter, outliers will be defined as those data points with values greater than three standard deviations from the mean of the sample. Data points identified as outliers will be examined further to correct for possible transcription and data entry errors but will be omitted from the analysis if no such administrative errors exist.

(iii) Analysis

The project proponent will maintain project database that includes the following data that can be directly attributable to each batch within the project, thereby allowing unambiguous determination of the emission reductions attributable the project:

- A list of households participating in project activity including name, community/location, distribution/installation date and unique serial number;
- Where replacements are made, assurance that the efficiency of the new ICS is similar to the specified.

Data obtained from the samples will be used to estimate proportions and mean values for the parameters described above. The values will then be factored into/applied for the emissions reduction calculations and that will further result in the request for issuance of VCUs. The project stoves/ICS that are not in use will be excluded from emissions reductions calculations and will not be counted towards the total number of ICS in operation during the monitoring period.

(iv) Data archiving

Soft copies of original stove purchase contracts, information collected at the time of registration or other means of acceptance by the users will be stored in the project office. Data base of surveys will be kept with a back-up and will be stored on an electronic medium. All data monitored and required for verification and issuance will be kept for two years after the end of the crediting period or the last issuance of credits for the project activity, whichever is later.

(c) Implementation

Sampling for the purpose of emission reduction calculation and elaboration of the monitoring report will occur at the end of each monitoring period.

- -sampling will be conducted by trained personnel from project proponent or an experienced third-party entity.
- -the maximum length of one monitoring period will be two years (duration, not calendar years), with option for annual or bi-annual monitoring period.
- The project proponent/Implementer will be responsible for household data collection and entry into the project database.

Field personnel will receive training on how to properly deal with surveying techniques and reduce errors and sign a document certifying that there is no conflict of interest of those involved in data collection and analysis. If there is conflict of interest, the personnel will not be allowed to participate in data collection and analysis. Appropriate data archiving methods will be adapted.

Project database will be able to determine the current status of project, the duration of previous monitoring periods, the monitoring data, and current verification activities.

(i) Assessment for Leakage

According to methodology VMR0006, version 1.1, section 8.3; a net to gross adjustment factor of 0.95 considered as default for accounting leakage. Hence the surveys are not required to determine leakage.

The other source of leakage occurs if equipment currently being utilized is transferred from outside the boundary to the project activity. All ICS in the project will be newly manufactured/assembled or newly installed.

(ii) Disposal of Low Efficiency Appliances and Use of Baseline Stoves

When an ICS is installed, the end user receives information explaining that conventional open fire appliance must no longer be used. Follow-up meetings with end users will ensure that those who have received an ICS are using it properly and that the conventional open fire is no longer in use. If it is determined that the conventional open fire is still in use along with the operating ICS, a survey will be conducted, a survey will be conducted to understand the reasons for the switch.

Monitoring Reporting

The project proponent will assess all monitoring data and produce a monitoring report for the VVB to verify corresponding to the preceding monitoring period. This report will present the data relating to the emission reductions generated by the project activity during the monitoring period.

APPENDIX 1

1. Public notice on 14/01/2023 for stakeholders' consultation meeting in the local new newspaper

SATURDAY | JANUARY 14, 2023

THE PHILIPPINE STAR
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departure time to 7:10 in the morning. The flag carrier plans to keep the morning timetable for its Guangzhou trips to offer passengers early day arrivals.

Further, PAL said it would deploy an Airbus A330-330 with up to 363 seats and an Airbus A321neo with a maximum of 199 seats for its Manila to

projects are expected to be implemented within this crop year," VMC said.

For its raw sugar and molasses operations, VMC operates a raw sugar mill with a daily capacity of 15,000 metric tons (MT).

It also operates a refinery plant with a daily capacity of 25,000 50-kilogram bags and provides toll refinery services to traders and planters for their raw sugar milled by other sugar centrals to ensure maximum utilization of the refinery.

The distillery operations produce alcohol and ethanol with an actual daily capacity of 120,000 liters with molasses as the primary raw material, which is sourced from sugar operations which produce it as a by-product.

LOCAL STAKEHOLDER CONSULTATION MEETING "INSTALLATION OF HIGH EFFICIENCY WOOD BURNING COOKSTOVES IN PHILIPPINES"

C-Quest Capital is developing a project titled "Installation of High Efficiency Wood Burning Cookstoves in Philippines" under Verified Carbon Standard (VCS) and the Sustainable Development Verified Impact Standard (SD VISTA). The project aims to provide clean energy access to households through the distribution of fuel-efficient cookstoves in rural and semi urban communities of the Philippines.

The project will contribute to the sustainable development goals through decreased demand for wood fuel, reduced deforestation, reduced drudgery in wood gathering and cooking time, increased skills and employment for the local community, and improved health through reduced households air pollution.

In this regard, C-Quest Capital extends an invitation to all stakeholders as well as interested parties for participating in the Local Stakeholder Consultation scheduled to take place on February 14, 2023 at J Leadership Center, #7 Water Lily St., San Nicolas, Floridablanca, Pampanga to discuss the project design, its effect on sustainable development as well as to take your valuable feedback.

For registration, please contact us on +639161851718 or write to us at: cqc_csat@cquestcapital.com. You may also send in your queries or comments at the above-mentioned address. The detail description of the project can be referred from the link shorturl.at/CIX19

2. Sample email invitation sent from C-quest on 24/01/2023

CQC_CSAT

To: Bhaskar Jyoti Nath; Vijay Kumar Machcha; CQC_CSAT

Bcc: bishoprudyjuan@gmail.com; gmscheuerman@gmail.com; jeffscheuerman@yahoo.com; a2rondc@yahoo.com; maxmaregmen@yahoo.com; ambrociolabtan26@yahoo.com; jorgee.rcc@gmail.com; edarosef@yahoo.com; rachel.alcantara@deped.gov.ph; gerrygalalcantara@gmail.com; Alvinmirandoselmo@gmail.com; revn oel2004@gmail.com; gfdelacruz@philrice.gov.ph; rolando.fredeluces@deped.gov.ph; mercedes.delacruz002 @deped.gov.ph; cndc112010@gmail.com; Remediosamarles@gmail.com; celbiteng@gmail.com; revjcm_23 @yahoo.com; iris_picardal@yahoo.com; rdungao4@gmail.com; jaysondomingo1977@gmail.com; noytalisc @yahoo.com; wilsonladringan@yahoo.com; revdrjonjiedelaraosa@yahoo.com; pesoporac@yahoo.com; marie |eilanieela@gmail.com; dejesuspol@gmail.com; dodie_mallari827@yahoo.com; edengarcia552@yahoo.com; elmer victoria101@gmail.com; lenymeman@eamcb.com; heeyoungpark256@gmail.com; alejandro_tin@yahoo .com; elvielucion30@gmail.com; louiecasupanan29@gmail.com; ptrjessalvarado@gmail.com; mirasoldavid3 5@gmail.com; pilarisipsantos@gmail.com; ritaocampo0226@gmail.com; mswdfloridablanca@gmail.com; ms wdfioridablanca@gmail.com; mswdfloridablanca@gmail.com; juniefer.bernardo001@deped.gov.ph; kenn200 42003@gmail.com; heddypituc2703@gmail.com; lutanmarty@gmail.com; coach21@gmail.com; aaeliangco @yahoo.com; payanayruel123@gmail.com; christinedavid0521@gmail.com; bertro112@gmail.com; cp_abras aldo11782@yahoo.com; dennisestrosos@gmail.com; zitroarlene08@gmail.com; mannycunanan99@gmail.co m; rs1254188@gmail.com; julietasantos0303@gmail.com; nerissa_gania@yahoo.com; Roxasdanilo00@gmail .com; henryvicente42@yahoo.com; owel73@gmail.com; ronelvallejo@gmail.com; williamdacusin@gmail.com ; armando34581@gmail.com; joebertlugtu36@gmail.com; mallarijaysonandrew@gmail.com; tuazonjayagusti n27@gmail.com; natividadlundang21@gmail.com; glendabernardo712@gmail.com; ram03@gmail.com; ram 03@gmail.com; elizarmann06@gmail.com; ncarreon561@gmail.com; fadinoricardo07@gmail.com; dimlafroil anb@gmail.com; janemagsino7@gmail.com

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Tue 1/24/2023 1:26 PM


Project summary and agenda... 141 KB


Non-technical summary - Phi... 1 MB

2 attachments (1 MB) [Save all to OneDrive - C-Quest Capital](#) [Download all](#)

Dear Sir / Madam,

C-Quest Capital, as part of the requirement under the process, is pleased to invite you to participate in the consultation meeting of Local Stakeholders for the project entitled 'Installation of high efficiency wood burning cookstoves in Philippines' scheduled on February 14, 2023, at 9: 00 AM in J Leadership Center, #7 Water Lily St., San Nicolas, Floridablanca, Pampanga (45 minutes' drive from Clark Airport). Interested stakeholders and parties can also join the consultation through the link or phone numbers below –

Video call link: <https://meet.google.com/azs-fdhq-xue>
 Phone numbers: <https://tel.meet/azs-fdhq-xue?pin=9152972117405>

The project aims to provide clean energy access to households through the distribution of fuel-efficient cookstoves in rural and semi urban communities of the Philippines. The project will contribute to the sustainable development goals through decreased demand for wood fuel, reduced deforestation, reduced drudgery in wood gathering and cooking time, increased skills and employment for the local community, and improved health through reduced household air pollution.

Considering the financial implications associated with the project, carbon finance will be used to increase the feasibility and scale-up of this project. Carbon Finance will be used to increase the feasibility of this project as it will help in providing clean energy access to households that lack awareness, required resources, and technical know-how.

The proposed project will be developed under the Verified Carbon Standard (VCS) and the Sustainable Development Verified Impact Standard (SD VISTA) to obtain SD VISTA labelled Verified Carbon Units (VCUs). Carbon credits will be used towards providing access to clean cooking technology which is currently deficient in the target communities owing to lack of awareness, required resources, and technical know-how.

A detailed agenda of the event with a non-technical project summary are attached to the email for your consideration. The detail description of the project activity can be referred from the link <https://shorturl.at/CIX19>

In case of any clarification or comment/suggestion regarding said projects, you can contact us at cqc_csat@cquestcapital.com

We look forward to your participation.

3. Stakeholders' Meeting Q&A

To make this Stakeholders Consultation meeting interactive, a question-and-answer session was facilitated to enable attendees to clarify their queries about the programme and gave the panelists a chance to explain various aspects of the programme. Stakeholders' comments and responses have been summarized below.

Name	Stakeholder Comment	Response (by CQC)
Bishop Rodolfo Juan	On behalf of the Filipino United Methodist and those of us joining you here virtually, we would like to say thank you, to you and to your team. Thank you so much for coming to our country to help us mitigate the negative result of climate change and to protect our environment. My question is where will these stoves be manufactured and how much are we going to pay for one piece? That's my simple question	We are importing these stoves actually from China, there is a manufacturing company called SSM and they are into producing the most efficient ones ,(Also briefed about the manufacturer technical specifications of stoves) The stoves that are being distributed under the project activity are at zero cost.
Pastor Arnel Lianco	Will you give to the small business owner like someone who has a small canteen or store?	Response 1: The present project activity is distribution of improved cookstoves to households and not for small business owners. It's a voluntary programme and not commercial purpose. Response 2: In the future we will be planning to also cater to business or commercial purposes like small tea rooms/restaurant which are still experimenting stage. Response 3: Our goal is to lessen the use of firewood, if we give the

		stoves to those who are already using Gasoline/gas then that means we are adding more people to use firewood and so we will not achieve our key goal.
Ptr. William Ladrigan	When will the stoves be shipped and how many are the target beneficiaries?	Response 1: Stoves already arrived; 250,000 households were targeted to be distributed within 3 years initially and distribution will start the next day (15th Feb 2023) We already have a list of beneficiaries for 4,000 households from West Pampanga and Zambales, San Marcelino and Castillejos and we are now processing the stoves for Tarlac and Bulacan.
Rev. Danilo Garcia	Do you have a memorandum of agreement for the recipients/beneficiaries?	We have an MOA Or an end-user agreement signed with each of the beneficiaries. For the beneficiaries of the pilot project we told them to be accountable to each other. We have (IP) already given some tents and then they just sold them. Why? Maybe because they have the product out of gratitude or maybe they want to buy alcohol and they don't have money. That is just an example of our personal experience. That is why please take care of the stoves well, if there is an issue, let us talk about

		it. Please do not sell or give away the stoves.
Rev. Rommel Vallejo	What type of work can the community avail and how many people will you hire?	Explained the 3 job possibilities: during Pre-distribution, distribution, and post-distribution.
Rev. Rommel Vallejo	How do you have a satellite office here in Pampanga or nearby community?	Response 1: C-Quest is in partnership with 3D Ecolutions and Lamuwan in the Philippines. Response 2: our office is in San Ramon, Lamuwan Environmental Management Consultancy Services registered under the Department of Trade and Industry and Municipality of Floridablanca, Pampanga and at present we have 4 Agents who are arranging everything. She also thanked the pastors and coordinators like DS Rev. Marty and the people of San Marcelino and Castillejos for their help.
Engr. Mangune	Are the stoves patented? Is it okay if other people want to manufacture the same?	It was clearly informed that the stoves are patented by SSM and must not be copied or reproduced.
Rev. Danilo Garcia	Are you going to show a demo today? On how to use it?	The demo of installation and operation will be tomorrow at ECC during the actual distribution. She also explained that there is a manual at the boxes on how to assemble and use, the do's and the don'ts with respect to improved cook stoves.

Rev. Danilo Roxas of Bulacan	We are so thankful for the many stoves that we will be receiving in which we will be using firewood and the source of firewood is a Tree. With lots of stoves then we will be using lots of firewood and lots of trees will be cut. Because we can only get firewood from a tree. Such can cause floods and soil erosion. I am afraid that if you will give 100,000 stoves which can be used for 10 years then it will also mean cutting off many trees. Do you have a program about it?	Response 1: Leslie explained that the beneficiaries are those who are using traditional cookstoves using firewood and not those using Gasol/gas stoves. Response 2: Putting it in the right perspective if at present you are using 10 kg firewood, with the new stove you will only use 5 firewood. So if now you are getting firewood from 100 trees using the new stove only 50 trees will be used.
Ptr. Arnel Liangco	As a coordinator he met the prospective beneficiaries and share the Gospel and the benefits of the project. The question is, "how about those who are renting an apartment and not permanent residents are they qualified to be beneficiaries?"	Yes. if they are presently using firewood then they are qualified. Granted that they will report to us if they will be transferring from one place to another. The project is expected to be implemented nationwide.
Ptr. Arnel	What will be the consequences like what if the stove is being stolen?	Please report it ASAP the hotline number or to any available staff, for us to report it to C Quest. She encouraged open communication with each other since it is a partnership project. Vijay added to please take care of the stoves and don't let them be lost or stolen for they will greatly benefit from it.
Rev. Ruel B. Payanay:	What if in one house there are 3 or more families living together? How many stoves will they be receiving?	Response 1: As of now, we will be providing two stoves per household as a standard. If the families are cooking separately in separate kitchens two stoves per

		family would be considered based on observation of the kitchen and household condition.
Mrs. Rose Ladrangan	Q1: We have a children's home, and we are feeding many children, how many stoves can you give to us? We are using firewood; the traditional stove and I use 4 Stoves to feed my children.	Response 1: We are providing two stoves per household as a standard practice of distribution. As these are highly efficient stoves, they could suffice the requirements of the four baseline stoves. However if there is any issue you can reach out us through helpline channels.
Rev. Jestril Alvarado	will you be able to cater to establishments like schools or institutions that are serving the community?	Our beneficiaries are primarily those who are using firewood for cooking in HHs and not for small business owners. It's a voluntary programme and not commercial purpose. The stoves are for small containers or cookware and not designed for big bowls. Like as Vijay said before in the future we will be planning to also cater to business or they are working on something for commercial use or institutions. Response 2 : we will be giving 2 stoves to each household. You can share your opinions and whatever feedback even with the designs and anything so we can improve our service.
Rev. Owell Lagman:	What is the difference of the two stoves?	CQC briefed about the two stoves once again and explained the specifications of the stoves.

		Along with presenting cookstoves photos on the screen.
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4) Stakeholders' consultation meeting photos

